

LATENTSPRING: PHYSICS-INFORMED MOLECULAR FLOW MATCHING FROM ATOMIC COMPOSITION

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ABSTRACT

Generating a molecule from its atomic composition requires the model to infer spatial organization without a supplied chemical graph. We introduce LatentSpring, a flow-matching framework whose source distribution is a mixture of harmonic networks over latent trees. Gaussian edge displacements encode atomic length scales, and a matrix-tree expression gives the normalized density after the tree is marginalized. The source works with an equivariant molecular flow and requires no bond labels or learned source network. In matched OMol25 fine-tuning experiments, it improves raw chemical-graph support over an isotropic Gaussian source by 5.86, 8.83 and 8.20 percentage points on panels of 12, 10 and 24 neutral organic compositions, respectively. A calibrated Gaussian covariance control supports a role for mixture structure beyond second moments. We also introduce physical feedback through paired training targets, reducing raw energy and force in both model continuations under eSEN and independent GFN2-xTB evaluation. A separate molecular torsion experiment verifies complete escorted-work reweighting against numerically resolved local targets. LatentSpring combines these source and training choices in a single neural generator, using 128 denoiser evaluations per sample without energy-based selection or geometry optimization at inference.

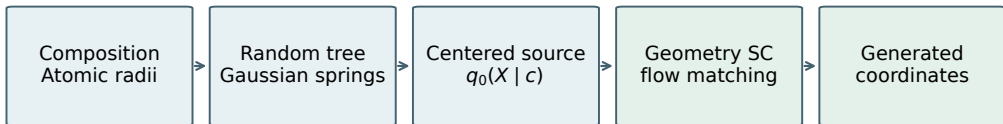
1 INTRODUCTION

Molecular generators must turn an unstructured coordinate distribution into a configuration compatible with its atoms. The source distribution determines part of this learning problem: independent Gaussian coordinates carry little molecular organization, while graph-conditioned harmonic priors encode relationships that must already be known. This distinction matters when the input specifies atomic composition and electronic conditions without a chemical graph, as in the bond-free OMol25 setting considered here.

LatentSpring introduces spatial dependence before learning the transport, while leaving the final chemical graph unspecified. We sample a latent tree over atoms, draw Gaussian displacements along its edges at atom-dependent scales, integrate them into coordinates, and remove translation. Marginalizing the tree defines a mixture of harmonic networks. The tree is an auxiliary source variable: the neural field receives coordinates and molecular conditions, and can generate chemical connectivities different from the sampled source tree. A matrix-tree expression makes the mixture density and its posterior-averaged elastic coupling explicit.

We integrate this source with equivariant flow matching (Lipman et al., 2023) and geometry self-conditioning (SC). Controlled comparisons hold the backbone, initialization, training data, optimization and inference budget fixed while changing the source. The harmonic mixture improves raw graph support in both continuations and repeats its advantage on two subsequent composition panels. Comparing with both a narrower shell source and an approximate covariance-matched Gaussian distinguishes the useful source structure from simply imposing near-fixed distances or increasing second moments.

Physical feedback provides a second way to improve the generator. We construct local training samples using forces at generated anchors, fit a physical-target model and a matched replay model, and transfer the difference between their parameter updates. This brings physical information into the network itself. For energy reweighting, known local reference densities make the complete



Source trees are latent dependence structures; no supplied chemical bonds

Figure 1: LatentSpring generation. Atomic radii and species propensities define a random harmonic network for drawing initial coordinates. The neural field takes these coordinates and the atomic conditions as input, and learns to generate the final molecular geometry. Physical feedback enters through training.

Jarzynski-type work tractable. We test the resulting generator on new compositions and examine the work correction separately on molecular torsional targets.

Our contributions are: (i) a normalized molecular harmonic-mixture source with latent spatial connectivity and no supplied bond graph; (ii) its integration with molecular flow matching and training-only physical feedback; and (iii) controlled source comparisons, tests on new compositions and evaluation under two energy models. The generator experiments measure structural support and energy/force scores at raw coordinates. The work experiments test distribution recovery against explicit local reference targets.

2 RELATED WORK

Molecular flows and source distributions. FlowMol3 combines equivariant coordinate updates with self-conditioning, fake atoms and geometry distortion (Dunn & Koes, 2026); its vector field forms our backbone. ET-Flow uses harmonic sources constructed from a supplied molecular graph (Hassan et al., 2024). More general source adaptation includes guidance under a frozen field (Wang et al., 2026) and learned conditional sources (Kim et al., 2026). LatentSpring constructs its molecular source from atomic composition by integrating over auxiliary spatial trees.

Distributions over trees. Factored tree distributions and tractable tree learning provide the probabilistic basis for our construction (Meila & Jaakkola, 2000; Duan & Dunson, 2023). Gaussian mixtures over random spanning-tree differences have also been used in imaging inverse problems (Everink, 2026). Here, isotropic edge vectors and atomic length scales define a density on centered molecular coordinates. Our Gaussian, shell and covariance controls test which aspects of this structure help the molecular flow.

Physical feedback and nonequilibrium work. OMol25 provides molecular structures, energies and forces across electronic conditions (Levine et al., 2025). Energy-weighted flow matching (Dern et al., 2025) and target-density training with importance sampling (Midgley et al., 2023) incorporate physical information into generative models. Physical-reward fine-tuning offers a further route for molecular diffusion models (Zhou et al., 2025). Jarzynski’s equality relates nonequilibrium work to equilibrium free-energy differences; Hummer–Szabo reconstruction applies it to single-molecule pulling (Jarzynski, 1997; Hummer & Szabo, 2001). Escorted transport extends this relation to imposed flow fields with the corresponding volume correction (Vaikuntanathan & Jarzynski, 2008). We use normalized local references to specify the physical training targets and their complete work weights.

3 A HARMONIC SOURCE WITH LATENT CONNECTIVITY

Let $c = (z_1, \dots, z_N, Q, S)$ contain atomic numbers, total charge and spin multiplicity. Coordinates lie in $\mathcal{H}_N = \{X \in \mathbb{R}^{N \times 3} : \sum_i x_i = 0\}$, with intrinsic Lebesgue measure and dimension $3(N - 1)$.

For positive symmetric affinities a_{ij} , define

$$p(T | c) = \frac{\prod_{(i,j) \in T} a_{ij}}{\tau(a)}, \quad \tau(a) = \sum_{T \text{ spanning tree}} \prod_{(i,j) \in T} a_{ij}. \quad (1)$$

We use factorized affinities $a_{ij} = u_i u_j$, with $u_i = 0.05$ for H and F/Cl/Br/I, and otherwise $u_i = \max(1, \text{default valence}(z_i) - 1)$. These propensities favor low degrees for terminal species such as hydrogen and halogens. Affinities depend on atomic species; charge and spin are checked for admissibility.

Gaussian edge laws and sampling. Let ℓ_{ij} be the sum of covalent radii and $\sigma_{ij}^2 = \ell_{ij}^2 \exp(s^2)/3$, with $s = 0.2$ fixed throughout. The edge-displacement law is

$$h_{ij}(d) = (2\pi\sigma_{ij}^2)^{-3/2} \exp\left[-\frac{\|d\|^2}{2\sigma_{ij}^2}\right]. \quad (2)$$

This scale matches the edge covariance of the lognormal shell control. We sample a tree with the weighted Wilson algorithm, draw its independent edge vectors, integrate along the tree, and subtract the coordinate mean. Both the sampling law and density are invariant under rotations and permutations of identical species.

Proposition 1 (Intrinsic mixture density). *For normalized isotropic edge densities,*

$$q_0(X | c) = N^{3/2} \frac{\tau(a_{ij} h_{ij}(x_i - x_j))}{\tau(a)} \quad (3)$$

is normalized on $\mathcal{H}_N$. For a log affinity $\alpha_e = \log a_e$,

$$\partial_{\alpha_e} \log q_0(X | c) = \Pr(e \in T | X, c) - \Pr(e \in T | c). \quad (4)$$

The proof is given in Appendix C. Two grounded Laplacian determinants evaluate the tree sums in cubic time, using positive log-space elimination for numerical stability. Equation (3) therefore gives the marginal source density without enumerating trees.

Elastic coupling. For a fixed tree, define the dimensionless spring energy

$$\mathcal{U}_T(X) = \frac{1}{2} \sum_{(i,j) \in T} \frac{\|x_i - x_j\|^2}{\sigma_{ij}^2}. \quad (5)$$

The component is a normalized Gaussian proportional to $e^{-\mathcal{U}_T}$. The marginal $-\log q_0$ is a soft mixture of these spring energies, with the component normalization constants retained. Its coordinate score is

$$\nabla_{x_i} \log q_0(X | c) = - \sum_{j \neq i} \Pr((i,j) \in T | X, c) \frac{x_i - x_j}{\sigma_{ij}^2}. \quad (6)$$

Differentiating Equation (3) gives this posterior-averaged elastic coupling, which describes how the marginalized source organizes atomic positions. In this auxiliary energy, springs have zero rest displacement and covalent radii set the scale of spatial fluctuations. Sampling uses direct draws from the harmonic mixture; Equation (6) describes its elastic structure.

4 INTEGRATION WITH MOLECULAR FLOW MATCHING

Sample $X_0 \sim q_0(\cdot | c)$ and a reference X_1 from the training corpus. A type-preserving assignment and proper rotation couple the pair, followed by shared independent Haar rotation and within-label permutation augmentation. This preserves the invariant source marginal while allowing dependence between paired endpoints (Appendix D). We use the linear path $X_t = (1-t)X_0 + tX_1$ and target $V^* = X_1 - X_0$.

We parameterize velocity as $v_\theta = D_\theta - X_t$, where $D_\theta(X_t, t)$ is the output of the FlowMol coordinate updater. A first pass yields $v^{(0)}$ and the endpoint estimate $\hat{X}_1 = X_t + (1-t)v^{(0)}$. A second pass

recycles that estimate through the existing geometry self-conditioning modules to obtain $v^{(1)}$. Both passes remain differentiable. Averaging coordinate errors within each molecule gives

$$\mathcal{L}_{\text{FM}} = \frac{1}{2} \mathbb{E} \left[\frac{\|v^{(0)} - V^*\|_F^2 + \|v^{(1)} - V^*\|_F^2}{3N} \right]. \quad (7)$$

Atomic categories and charge inputs remain fixed, and edge inputs carry a constant no-bond state. All primary source comparisons share this coordinate-only objective and the same two-pass self-conditioning architecture.

Generation integrates $v^{(1)}$ with 32 midpoint steps and two passes per field call, totaling 128 primitive denoiser evaluations per attempted sample, followed by 0.025 Å centered Gaussian noise. All source comparisons use this same sampler. We evaluate its raw coordinates directly, without subsequent energy selection or geometry optimization.

4.1 PHYSICAL FEEDBACK THROUGH PAIRED TRAINING TARGETS

A frozen harmonic generator supplies training anchors A on compositions excluded from evaluation. The teacher uses parity-averaged eSEN energies $E_+(X) = [E(X) + E(-X)]/2$ and forces $F_+(X) = [F(X) - F(-X)]/2$. For graph-supported anchors, a bounded force displacement defines local escorted samples $Y = X + \delta_A$, with $X \sim q_A = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. The local scale and displacement are fixed from the anchor force before drawing particles (Appendix G). We train a physical-target continuation and a replay continuation from the same weights and matched training draws, then transfer their parameter difference:

$$\theta_* = \theta_0 + (\theta_{\text{physical}} - \theta_{\text{replay}}). \quad (8)$$

This parameter-arithmetic update (Ilharco et al., 2023) uses coefficient one and leaves model buffers fixed. Physical information is thus carried by the trained network during generation.

Complete escorted-work weighting (Vaikuntanathan & Jarzynski, 2008) defines a local physical extension. On its specified structural support, let

$$W_A = E_+(Y) - E_+(A) + kT[\log q_A(X) - \log q_A(Y)]. \quad (9)$$

The reference-density terms account for the escort’s shift in proposal law. The weight $\exp(-W_A/kT)$ estimates the local energy-tilt normalizer. We compare uniform and complete-work training weights, and use the less expensive force update as the primary candidate for the new-composition physical evaluation. Appendix G gives the target definition, particle construction and training controls.

5 EXPERIMENTAL DESIGN

Task and training. We study adaptation of a pretrained molecular flow to coordinate generation from neutral singlet organic compositions. Each composition contains C and H and uses elements from H/B/C/N/O/F/Si/P/S/Cl/Br/I. Experiments cover 8–40 atoms; the implementation supports up to 200. Two continuations start from a shared pretrained checkpoint. Each uses 3,000 distinct OMol25 structures and 3,000 optimizer updates, with identical data and initialization across its source variants. The vector field has 5,904,369 parameters. The backbone learning rate is $2 \cdot 10^{-5}$ and the self-conditioning modules use $3 \cdot 10^{-4}$.

Training structures pass the graph assay and a geometric filter requiring a bounded-edge, degree-capped scaffold. This filter is specified in the data-selection protocol. The models receive the selected coordinates; the scaffold is used for selection only. All source variants use the same filtered subset.

Composition panels. Reference qualification accepts 31 of 78 metadata-eligible compositions. Of these, 21 had appeared in earlier development studies. We use a fixed 12-composition subset for the source comparison and all 10 remaining compositions for subsequent confirmation. The latter have 17–28 atoms and are absent, by exact composition count-vector checks, from 3,902,107 processed training rows and 39,415 validation rows. Reference geometries serve only to qualify the evaluation conditions. Both panels use the same two fitted continuations. A further 24-composition test is described in Section 6.1.

Table 1: Chemical-graph support per attempted output. All primary arms share the geometry-SC backbone and 128 primitive denoiser calls. The additional panel uses frozen models selected after the development comparison.

Panel	Continuation	Gaussian	Shell tree	Harmonic tree
Development (12)	1	357/768	417/768	436/768
	2	377/768	342/768	388/768
Additional (10)	1	326/640	—	413/640
	2	349/640	—	375/640

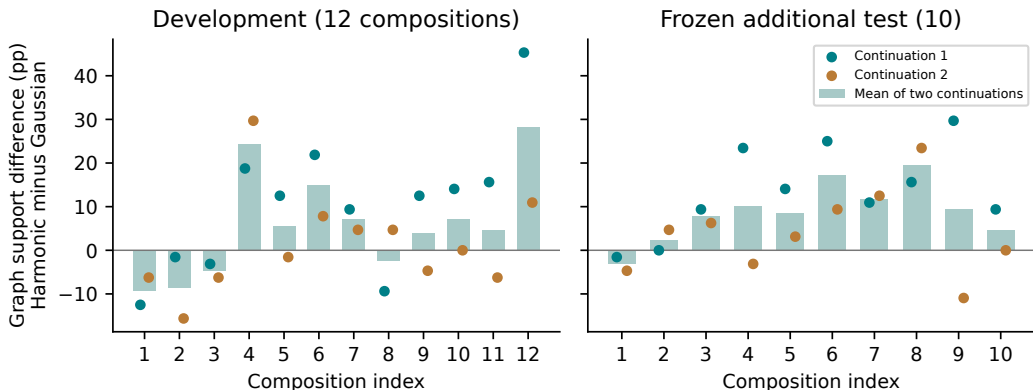


Figure 2: Harmonic-source graph-support gains vary across compositions and continuations. Points show individual continuation differences; bars average them. Both panels evaluate the same two trained models.

Metrics and uncertainty. The first two panels use 64 samples per composition and the 24-composition test uses 32. Geometry support requires a connected contact graph and no short-distance overlap, with thresholds of 1.25 and 0.6 times the summed covalent radii. Graph support additionally requires RDKit to construct a connected, sanitized graph with the specified total charge. We report support per attempted output, including failed samples and validator exceptions in the denominator. Distinct perceived connectivities within each composition measure graph diversity; timings record the full generation procedure. Appendix A gives the selection and perception settings.

Unless stated otherwise, 95% confidence intervals use paired bootstrap resampling of generation draws, holding the fitted models and compositions fixed. We also bootstrap compositions to describe variation across the evaluation panel. Physical scores are computed at the saved raw coordinates using the declared electronic conditions and approximate potentials.

6 RESULTS

Harmonic trees improve graph support over isotropic Gaussian sources by 5.86 percentage points on the development panel (95% CI [2.67,8.98]; composition interval [0.52,12.17]). Both continuations improve, with a larger gain in the first. The harmonic source also outperforms the narrow shell by 4.23 points; the shell-minus-harmonic interval is [-7.29,-1.11].

On the 10-composition confirmation panel, harmonic sources raise graph support from 52.73% to 61.56%, a gain of 8.83 points (95% CI [5.08,12.50]; composition interval [4.84,12.66]). Geometry support improves by 9.69 points [6.33,12.89]. Every graph-supported output has distinct perceived connectivity within its composition, and the evaluator encounters no exceptions. Generation takes nearly the same time for the two sources: harmonic/Gaussian timings are 175.73/175.43 seconds and 178.20/177.74 seconds for 640 samples in the two continuations.

Table 2: Covariance-Gaussian control at matched continuation and inference budgets, using the calibrated 128-tree covariance estimate.

Panel	Continuation	Covariance Gaussian	Harmonic tree
Development	1	332/768	436/768
	2	313/768	388/768
Additional	1	286/640	413/640
	2	305/640	375/640

Table 3: Fresh 24-composition graph support, 768 attempts per continuation. All rates describe raw outputs before any structural relaxation. All methods use 128 primitive denoiser calls per attempted output.

Model	Continuation 1	Continuation 2	Pooled rate
Gaussian reference	332	352	44.53%
LatentSpring harmonic	435	375	52.73%
With force update	439	382	53.45%
With complete-work update	426	383	52.67%

Control for source covariance. To test the role of second moments, we construct a Gaussian from a 128-tree estimate of the harmonic mixture covariance and symmetrize it over identical species. Calibration against an independent 8,192-tree estimate gives 1.85–6.25% relative Frobenius error across the 22 compositions. This control uses the same data, optimizer, network and sampling budgets. It is evaluated after the initial source comparison against the frozen harmonic checkpoints.

Harmonic sources improve graph support by 11.65 points on development (paired 95% [8.59,14.78]; descriptive composition [8.07,15.43]) and 15.39 points on the additional panel (paired 95% [11.64,19.06]; composition [11.95,18.98]). All four point differences are positive. The covariance Gaussian is also worse than the isotropic Gaussian in both panels. The harmonic mixture thus retains an advantage over this calibrated second-moment control, consistent with a useful contribution from its mixture structure. The comparison adds 2,816 outputs and holds parameter count, initial self-conditioning weights, training rows and optimization fixed.

Appendix B reports the additional panel’s eSEN scores and an original-checkpoint reference using endpoint-based sampling.

6.1 NEW COMPOSITIONS AND INDEPENDENT PHYSICAL EVALUATION

We next evaluate the frozen Gaussian, harmonic and two physical-update models on 24 new compositions. The force-only update is chosen as the primary physical candidate before this evaluation. Starting from a pool of 256 distinct compositions outside the earlier candidate sets, reference qualification accepts 121. Fixed metadata ranks select eight compositions from each of the 17–20, 21–24 and 25–28 atom bins. All 24 are absent from the processed OMol25 training and validation corpora. Reference coordinates are used solely for qualification. Each model generates 32 samples per composition in each continuation, giving 6,144 raw outputs in total.

The harmonic source improves graph support by 8.20 percentage points over Gaussian (95% CI [5.08, 11.33]), with gains in both continuations. The force update changes graph support by +0.72 points relative to harmonic, with interval [−0.98, 2.34], consistent with a similar support rate. Every accepted output has distinct perceived connectivity within its composition in these draws.

Energy and force scores. Every output is evaluated at fixed coordinates with eSEN and gas-phase GFN2-xTB single-point energies and gradients (Bannwarth et al., 2019). GFN2 provides an independent potential, used only for evaluation. Four of the 6,144 generated-geometry GFN2 calculations fail; all four concern graph-rejected outputs. Calculations succeed for every graph-supported output, and reference inversion checks pass. Including reference checks, evaluation costs 12,384 eSEN queries and 6,240 GFN2 attempts. eSEN scores use the parity averages defined in Section 4. Force RMS is the square root of the mean squared atom-force norm.

Table 4: Primary force-update change versus the frozen harmonic model on paired common-graph support. Both continuations improve under both potentials. Brackets give paired 95% confidence intervals.

Potential	Energy change (eV/atom)	Force-RMS change (eV/Å)
eSEN	-0.01308 [-0.01805, -0.00817]	-0.34793 [-0.41498, -0.28478]
GFN2	-0.01168 [-0.01616, -0.00724]	-0.35620 [-0.42983, -0.28714]

Table 5: Raw graph support of the independently adapted conditional EDM, 768 attempts per continuation. The harmonic and force-update models in Table 3 reach 52.73% and 53.45%, respectively, at 128 calls.

Sampler	Continuation 1	Continuation 2	Pooled rate
EDM, 128 calls	103	100	13.22%
EDM, 1,001 calls	93	106	12.96%

The primary contrast uses paired outputs that pass the graph assay for both models, with successful electronic calculations. Such pairs are available for all 24 compositions in both continuations. Energy and force improve under both potentials. The same direction holds when including all outputs for eSEN and all jointly successful calculations for GFN2. For the source-only harmonic–Gaussian comparison, aggregate energy also decreases, with a larger effect in one continuation than the other.

The complete-work and force-only updates perform similarly within sampling uncertainty. Their energy difference on common graph support is -0.00413 eV/atom under eSEN, with interval $[-0.00903, 0.00078]$, and -0.00363 under GFN2, with interval $[-0.00792, 0.00075]$. Force and graph differences also have intervals spanning zero. The force-only update therefore offers comparable physical scores with a lower teacher-preparation cost.

6.2 AN INDEPENDENTLY ADAPTED DIFFUSION BASELINE

We additionally adapt the official EGNN/EDM backbone (Hoogeboom et al., 2022) to the same composition-conditioned task. Positions follow its variance-preserving diffusion schedule, while clean atom categories are conditions during both training and sampling. The GEOM-pretrained checkpoint receives 30,000 updates on the same 3,000 OMol25 training structures used by each source continuation, with its original 10^{-4} learning rate. We evaluate both 128 denoiser calls and the full 1,001-call sampler, including the final observation step, on all 24 fresh compositions.

This comparison tests adaptation to the same task and training data with a different pretrained architecture and objective. EDM has 2.38 million parameters and one denoiser pass per update; LatentSpring has 5.90 million and two. The longer EDM continuation takes 481.51/463.61 seconds, compared with approximately 570–590 seconds for the source-model continuations. Its 128-call sampler takes 48.54/47.94 seconds per 768 outputs, and its 1,001-call sampler takes 377.65/374.12 seconds. These timings distinguish call count from computational cost. Pretraining histories differ, and composition-overlap checks cover the processed OMol25 corpora. The matched Gaussian–harmonic experiments provide the controlled comparison of source choice.

6.3 COMPLETE-WORK CORRECTION ON MOLECULAR ROTORS

To test the work correction directly, we select three methyl rotors from training molecules in fixed data order, using geometry alone. Each rotor defines a constrained one-dimensional molecular curve. A periodic interpolant of 512 GFN2 energies supplies the potential $U(\phi)$; the target is $\rho(\phi) \propto q_0(\phi) \exp[-(U(\phi) - U(0))/kT]$, where q_0 is a normalized von Mises reference of concentration one and kT corresponds to 300 K.

Starting from $\phi \sim q_0$, use escorts $\psi_a(\phi) = \phi + a \sin \phi$ for $a \in \{-0.5, 0, 0.5\}$. Their positive Jacobian is $J_a = 1 + a \cos \phi$. The complete generalized work includes both reference densities and $-kT \log J_a$ (Appendix H). We compare complete work, energy-only weights, weights omitting the

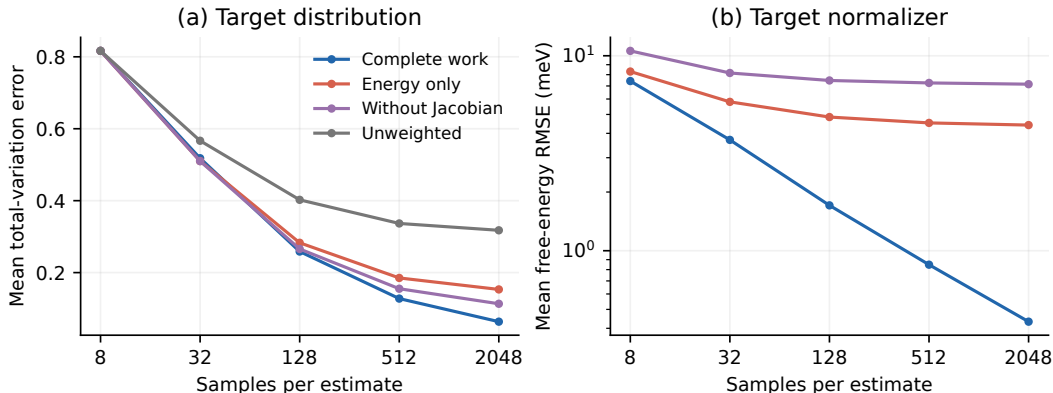


Figure 3: Complete work corrects the nonequilibrium sampling bias on the specified torsional targets. Curves average all three molecules and both nonidentity escorts; each case has 128 repetitions. Energy-only and incomplete-work controls retain bias as sampling increases. Unweighted endpoints have no free-energy estimator. Each target is a constrained molecular rotor at 300 K.

Jacobian, and unweighted endpoints, with shared draws and 128 repetitions at each of 8, 32, 128, 512 and 2,048 samples.

At 2,048 samples, complete work has lower mean total-variation error than both incomplete-weight controls in all six nonidentity cases. Across these cases, mean errors are 0.0636 for complete work, 0.1532 for energy-only weights and 0.1134 without the Jacobian, a 58.5% reduction relative to energy-only weighting. Complete-work free-energy RMSE ranges from 0.158 to 0.734 meV. Independent density-ratio quadrature also recovers the specified target normalizer in all nine molecule/escort cases. The identity-escort control makes the three weighted methods coincide. Together, the distribution and normalizer results show the role of complete work accounting on these molecular energy curves.

7 DISCUSSION

LatentSpring places molecular spatial structure in the source distribution, reducing the burden on the learned transport. The harmonic mixture improves graph support across all three panels, and the covariance control indicates that its benefit extends beyond the tested second-moment approximation. Physical training targets provide a complementary improvement in raw energy and force scores. The molecular rotor study explains why complete work accounting matters when reweighting an escorted sample distribution.

The experiments focus on a fixed adaptation budget and a selected neutral organic subset of OMol25. The two continuations share a pretrained checkpoint, so the reported intervals describe sampling variation conditional on those fitted models. Graph support is a structural criterion; the energy and force results describe the two approximate potentials at raw coordinates. The equilibrium targets tested here are the explicitly defined local ensembles. Broader chemical coverage, larger adaptation budgets and correction of full molecular ensembles are natural next tests of the framework.

REPRODUCIBILITY STATEMENT

The project includes code, model configurations and data-selection manifests. These specify random seeds, checkpoint hashes, evaluation settings and every attempted output. We report training, physical preparation and generation costs separately, and provide tests for source normalization, symmetry, density gradients and work identities. Source data and pretrained backbones retain their original licenses and access requirements.

AI USE STATEMENT

OpenAI Codex and Claude Code assisted literature review, hypothesis and experiment design, mathematical derivation and proof writing, code implementation, data preprocessing, testing, experimental analysis, and manuscript drafting and editing.

REFERENCES

- Christoph Bannwarth, Sebastian Ehlert, and Stefan Grimme. GFN2-xTB — an accurate and broadly parametrized self-consistent tight-binding quantum chemical method with multipole electrostatics and density-dependent dispersion contributions. *Journal of Chemical Theory and Computation*, 15(3):1652–1671, 2019. doi: 10.1021/acs.jctc.8b01176.
- Niclas Dern, Lennart Redl, Sebastian Pfister, Marcel Kollovich, David Lüdke, and Stephan Günnemann. Energy-weighted flow matching: Unlocking continuous normalizing flows for efficient and scalable boltzmann sampling. *arXiv preprint arXiv:2509.03726*, 2025. URL <https://arxiv.org/abs/2509.03726>.
- Leo L. Duan and David B. Dunson. Bayesian spanning tree: Estimating the backbone of the dependence graph. *Journal of Machine Learning Research*, 24(397):1–44, 2023. URL <https://www.jmlr.org/papers/v24/22-0252.html>.
- Ian Dunn and David R. Koes. FlowMol3: Flow matching for 3D de novo small-molecule generation. *Digital Discovery*, 5(5):2052–2066, 2026. doi: 10.1039/D5DD00363F. URL <https://doi.org/10.1039/D5DD00363F>.
- Jasper Marijn Everink. Random spanning tree markov random field priors for bayesian inverse problems in imaging. *arXiv preprint arXiv:2605.18619*, 2026. URL <https://arxiv.org/abs/2605.18619>.
- Majdi Hassan, Nikhil Shenoy, Jungyoon Lee, Hannes Stärk, Stephan Thaler, and Dominique Beaini. ET-Flow: Equivariant flow-matching for molecular conformer generation. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-4091. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/e8bd617e7dd0394ceadf37b4a7773179-Abstract-Conference.html.
- Emiel Hoogetboom, Víctor Garcia Satorras, Clément Vignac, and Max Welling. Equivariant diffusion for molecule generation in 3D. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 8867–8887, 2022. URL <https://proceedings.mlr.press/v162/hoogetboom22a.html>.
- Gerhard Hummer and Attila Szabo. Free energy reconstruction from nonequilibrium single-molecule pulling experiments. *Proceedings of the National Academy of Sciences*, 98(7):3658–3661, 2001. doi: 10.1073/pnas.071034098.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2212.04089>.
- Christopher Jarzynski. Equilibrium free energy differences from nonequilibrium measurements: a master equation approach. *Physical Review E*, 56:5018–5035, 1997. doi: 10.1103/PhysRevE.56.5018. URL <https://arxiv.org/abs/cond-mat/9707325>.
- Junwan Kim, Jiho Park, Seonghu Jeon, and Seungryong Kim. Better source, better flow: Learning condition-dependent source distribution for flow matching. *arXiv preprint arXiv:2602.05951*, 2026. URL <https://arxiv.org/abs/2602.05951>.
- Daniel S. Levine, Muhammed Shuaibi, Evan Walter Clark Spotte-Smith, Michael G. Taylor, Muhammad R. Hasyim, Kyle Michel, Ilyes Batatia, Gábor Csányi, Misko Dzamba, Peter Eastman, Nathan C. Frey, Xiang Fu, Vahe Gharakhanyan, Aditi S. Krishnapriyan, Joshua A. Rackers, Sanjeev Raja, Ammar Rizvi, Andrew S. Rosen, Zachary Ulissi, Santiago Vargas,

- C. Lawrence Zitnick, Samuel M. Blau, and Brandon M. Wood. The Open Molecules 2025 (OMol25) dataset, evaluations, and models. *arXiv preprint arXiv:2505.08762*, 2025. URL <https://arxiv.org/abs/2505.08762>.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://openreview.net/forum?id=PqvMRDCJT9t>.
- Marina Meila and Tommi Jaakkola. Tractable bayesian learning of tree belief networks. Technical Report CMU-RI-TR-00-15, Carnegie Mellon University, 2000. URL <https://publications.ri.cmu.edu/tractable-bayesian-learning-of-tree-belief-networks>.
- Laurence Illing Midgley, Vincent Stimper, Gregor N. C. Simm, Bernhard Schölkopf, and José Miguel Hernández-Lobato. Flow annealed importance sampling bootstrap. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://arxiv.org/abs/2208.01893>. arXiv:2208.01893.
- Suriyanarayanan Vaikuntanathan and Christopher Jarzynski. Escorted free energy simulations: Improving convergence by reducing dissipation. *Physical Review Letters*, 100:190601, 2008. doi: 10.1103/PhysRevLett.100.190601. URL <https://arxiv.org/abs/0804.3055>.
- Zifan Wang, Alice Harting, Matthieu Barreau, Michael M. Zavlanos, and Karl H. Johansson. Source-guided flow matching. In *International Conference on Learning Representations*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/adedc33539eafdf3d2c14a5e0032db6e-Abstract-Conference.html.
- Zhijian Zhou, Junyi An, Zongkai Liu, Yunfei Shi, Xuan Zhang, Fenglei Cao, Chao Qu, and Yuan Qi. Guiding diffusion models with reinforcement learning for stable molecule generation. *arXiv preprint arXiv:2508.16521*, 2025. URL <https://arxiv.org/abs/2508.16521>.

A DATA SELECTION AND STRUCTURAL EVALUATION

Training data are selected in a fixed shuffled order from the processed OMol25 training corpus, excluding the evaluation compositions. The first 3,000 eligible structures form each continuation’s training set. Eligibility requires the atom, element and electronic-state conditions in Section 5, the structural assay below, and a geometric scaffold filter.

The scaffold filter considers pairs whose distance divided by the sum of covalent radii lies strictly between 0.6501 and 1.1999. Kruskal’s algorithm constructs a minimum normalized-distance spanning tree from these pairs. A structure is accepted if the tree is connected and its degrees are at most 1 for H/F/Cl/Br/I, 4 for B/C/N/Si, 3 for O, and 6 for P/S. The scaffold is used only to select training coordinates; the molecular flow uses its own sampled harmonic source.

Structural evaluation uses RDKit 2025.03.2. After the contact-connectivity and overlap checks, DetermineBonds assigns bonds at the specified total charge, with `covFactor=1.25`, `useVdw=True`, `allowChargedFragments=True`, and `embedChiral=False`. Sanitization must succeed, the assigned formal charges must sum to the specified charge, and the perceived molecule must have one connected component. Graph diversity counts canonical SMILES after removing stereochemical tags and explicit hydrogens. Evaluator errors and unsupported outputs remain in the attempted-sample denominator.

B ADDITIONAL GENERATOR EVALUATIONS

Energy and force evaluation of the frozen generators. We score every output on the ten-composition panel and each original reference with the same conservative eSEN checkpoint at its original charge and spin. We use the parity averages $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$. Differences are paired within composition before averaging; energies per atom also subtract that composition’s reference energy, which cancels in the paired difference. There are 5,160 raw energy/force queries, including both inversions of the references in each continuation.

Table 6: Harmonic minus isotropic Gaussian physical readouts on the frozen additional panel. Brackets give paired 95% confidence intervals.

Readout	Support	Difference	Interval
Energy (eV/atom)	All attempts	−0.0406	[−0.0579, −0.0232]
	Common graph	−0.0340	[−0.0692, −0.0003]
Force RMS (eV/Å)	All attempts	−0.4635	[−0.6782, −0.2512]
	Common graph	−0.5389	[−0.8950, −0.2089]

Force RMS means the square root of the per-atom mean squared force norm, then averaged over samples. The common-graph statistic uses only paired outputs accepted by both methods, with 3–35 pairs per composition/continuation cell. All-attempt statistics also include graph-rejected outputs. We report both to separate the aggregate score from the score among jointly supported structures.

The aggregate improvement is concentrated in the first continuation. All-attempt energy differences are −0.08049 and −0.00070 eV/atom; force-RMS differences are −1.21996 and +0.29302 eV/Å, respectively.

Original endpoint-checkpoint reference. We evaluate the original 30,000-step OMol25 checkpoint on the coordinate-generation task by clamping composition and the neutral/no-bond inputs. Its endpoint sampler uses 127 Euler updates and a first-step bootstrap, totaling 128 denoiser calls. Retaining predicted categorical history gives 106/640 and 113/640 graph passes in two sampling streams. Fixing the history categories and centering the first bootstrap gives 116/640 and 120/640. Both streams use the same checkpoint; the second variant changes both categorical history and bootstrap centering. The LatentSpring source models additionally receive shared task adaptation and 3,000 specialized updates. We therefore use the matched Gaussian and covariance-Gaussian continuations to assess the effect of source choice.

C NORMALIZATION AND AFFINITY-GRADIENT PROOF

For a tree T , let $B_T \in \mathbb{R}^{(N-1) \times N}$ be an oriented incidence matrix and $U \in \mathbb{R}^{N \times (N-1)}$ an orthonormal basis of the centered subspace. Because $B_T \mathbf{1} = 0$, $(B_T U)(B_T U)^\top = B_T B_T^\top$. Cauchy–Binet expresses $\det(B_T B_T^\top)$ as the sum of squares of its N maximal minors. Each minor of a tree incidence matrix is ± 1 , so the determinant is N . The absolute Jacobian in one spatial dimension is therefore $\sqrt{N}$, and in three dimensions it is $N^{3/2}$. Consequently the component density is $q(X \mid T, c) = N^{3/2} \prod_{e \in T} h_e(B_T X)$ and integrates to one. Averaging with Equation (1) yields Equation (3) and preserves normalization. Differentiating the log of each finite positive tree partition with respect to α_e gives its edge-inclusion probability, proving Equation (4).

D SOURCE INVARIANCE UNDER SYMMETRY COUPLING

Let G be the compact group of proper rotations and permutations of identical conditioning labels. Suppose q_0 is G -invariant, $X \sim q_0$, and a pairing algorithm selects $g(X, Y) \in G$ using a target Y . If H is an independent Haar draw, then conditional on the orbit of X and on the selected g , HgX has the uniform orbit distribution of HX , by Haar invariance. Integrating over the unchanged source-orbit distribution recovers q_0 . This argument permits a non-Gaussian source and explains the shared augmentation used in the pairing. The paired endpoints remain correlated; our training objective uses their conditional displacement target directly.

E COVARIANCE CONTROLS

For $R = \ell \exp(sZ - s^2/2)$ with $Z \sim N(0, 1)$, $\mathbb{E}R^2 = \ell^2 \exp(s^2)$. A uniform direction has second moment $I/3$, so each edge has covariance $vI = \ell^2 \exp(s^2)I/3$. Let A_T map edge coordinates to centered node coordinates. Independence gives $C_T = A_T \text{diag}(v_e) A_T^\top$. Replacing the edge vectors by independent Gaussians of covariance $v_e I$ leaves C_T unchanged and, since all component

means are zero, leaves $\mathbb{E}_T C_T$ unchanged. The single-Gaussian control uses a finite, composition-seeded average of these matrices. Averaging this matrix over within-species permutations preserves positive definiteness on the centered subspace and yields species invariance. The Gaussian is exactly normalized for its chosen covariance; equality to the infinite tree-mixture covariance remains approximate.

F JARZYNSKI IDENTITY FOR ESCORTED TRANSPORT

Jarzynski’s relation connects exponential work averages with equilibrium free-energy differences, including in single-molecule pulling experiments (Jarzynski, 1997; Hummer & Szabo, 2001). Escorted transport introduces an imposed flow and includes its configuration-volume correction in the work (Vaikuntanathan & Jarzynski, 2008). The change-of-variables form provides the basis for our local-reference construction.

Fix a composition and intrinsic coordinate measure. Let $\rho_1(x) = \mathbf{1}_\Omega(x) \exp[-\beta E(x)]$ be a target with finite, nonzero $Z_1 = \int_{\mathcal{H}_N} \rho_1(x) dx$, for example a declared confined molecular ensemble. The confinement Ω is chosen to make the target normalizable. Let $q_0 > 0$ be the normalized source in Equation (3), and let $\Phi : \mathcal{H}_N \rightarrow \mathcal{H}_N$ be a differentiable bijection with nonzero intrinsic Jacobian. For $\Phi(x_0) \in \Omega$, define the generalized work

$$W_\Phi(x_0) = E(\Phi(x_0)) + \beta^{-1} \log q_0(x_0) - \beta^{-1} \log |\det D\Phi(x_0)|, \quad (10)$$

and set $W_\Phi = +\infty$ otherwise. Then

$$\mathbb{E}_{q_0}[e^{-\beta W_\Phi}] = \int_{\mathcal{H}_N} \rho_1(\Phi(x_0)) |\det D\Phi(x_0)| dx_0 \quad (11)$$

$$= Z_1. \quad (12)$$

Equivalently, the auxiliary source energy $E_0 = -\beta^{-1} \log q_0$ has $Z_0 = 1$ in the declared measure, yielding a Jarzynski-type normalizer identity. The Jacobian term accounts for compression by the escort map. The same complete weight estimates target observables through self-normalized importance sampling; its finite-sample variance depends on source–target overlap.

For an exact, sufficiently regular ODE flow, the intrinsic log determinant is the integral of the velocity divergence. Our implementation makes the work tractable through explicit local references: translations of Gaussian anchor distributions in Appendix G, and periodic rotor maps with analytic Jacobians in Appendix H. The neural generator itself is evaluated through its raw samples, using the midpoint solver and terminal noise specified in Section 4.

G LOCAL FORCE-ESCORTED WORK DISTILLATION

The physical teacher combines local force-guided proposals with energy-weighted flow matching (Dern et al., 2025; Midgley et al., 2023). Normalized anchor references make the escorted-work weights explicit (Vaikuntanathan & Jarzynski, 2008).

Local reference and target. For a generated training anchor A with accepted perceived graph G_A , define $q_A(X) = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. At $kT = 0.0258519998$ eV (300 K), remove net force and set

$$\sigma_A = \min \left(0.03 \text{ \AA}, \sqrt{\frac{0.10 \text{ \AA} kT}{\|F_+(A)\|}} \right), \quad \delta_A = \frac{\sigma_A^2}{kT} F_+(A). \quad (13)$$

A vanishing force selects the maximal width. These quantities are fixed before drawing $X \sim q_A$. The escort $Y = X + \delta_A$ is a translation on the centered subspace and has determinant one. The local unnormalized target is

$$\rho_A(Y) = q_A(Y) \exp \left[-\frac{E_+(Y) - E_+(A)}{kT} \right] \mathbf{1}_{G_A}(Y), \quad (14)$$

where the support requires the same labelled perceived bond orders, formal charges and radical counts, together with the connected/nonoverlap assay. The graph is inferred from the anchor coordinates. The resulting target is a local energy tilt restrained around A by q_A .

Table 7: Direct physics-distillation graph support per 640 attempts. The frozen reference is not refitted; other arms have identical additional update budgets.

	Continuation	Frozen	Replay	Unweighted escort	Complete work
1		418	340	330	344
2		377	345	371	362

Complete local work. With reference Hamiltonian $H_0 = -kT \log q_A$ and target Hamiltonian $H_1 = H_0 + E_+ - E_+(A)$ plus hard support, the generalized quench work is

$$W_A(X) = E_+(Y) - E_+(A) + kT[\log q_A(X) - \log q_A(Y)] \quad (15)$$

on valid support, and $+\infty$ otherwise. Change of variables gives $\mathbb{E}_{q_A}[\exp(-W_A/kT)] = \int \rho_A(Y) dY$. The two reference-density terms correct for the escort’s proposal shift. Analytic checks recover constant work for a constant-force escort and the normalizer and weighted mean for a quadratic energy.

Training and controls. Each continuation selects eight distinct 17–28-atom training compositions by a fixed metadata hash and generates 32 anchors per composition. These compositions are disjoint from evaluation. Invalid anchors remain in the generation counts and are excluded from the teacher pool. Each valid anchor produces eight escorted particles. Particles outside its declared support receive zero weight. The work arm samples training endpoints using self-normalized $\exp(-W_A/kT)$ weights; the unweighted escort arm samples the same valid particles uniformly. A replay arm uses the retained raw anchors, and a frozen model supplies the reference. All-zero-particle anchors are recorded and excluded identically from all student pools.

Each student alternates 500 original-reference FM updates with 500 generated-anchor endpoint updates, sharing initialization, anchor draws, optimization and sampling budgets. Physical preparation costs differ: replay needs no force queries, the unweighted escort needs anchor forces, and complete work additionally needs energies at valid proposed endpoints. Actual shared research cost and these method requirements are reported separately. The final generator uses the learned network alone. Its temperature feature stays fixed throughout training and evaluation; 300 K determines the local teacher weights. Self-normalization introduces finite-particle bias, and the reported teacher ESS is computed over the eight local particles at each anchor.

Direct-distillation outcome. Each continuation fits three students for 1,000 updates. Of 512 generated training anchors, 351 pass the graph assay and 349 have at least one supported escorted particle. Mean local particle ESS by composition is about 1.27–1.87 out of 8, indicating substantial weight concentration and limited finite-particle resolution of the local target.

On paired common-graph support, complete work versus the unweighted escort reduces mean energy by 0.00758 eV/atom (conditional paired 95% interval $[-0.01424, -0.00113]$ for the signed difference) and force RMS by 0.16174 eV/Å (interval $[-0.27660, -0.05590]$). Both continuations favor complete work. The all-output comparison has the same direction.

All three direct continuations reduce quality relative to the frozen model. Complete work has graph support 55.16%, versus 62.11% for frozen, and common-supported energy is *higher* by 0.10587 eV/atom (interval $[0.09009, 0.12168]$). Replay also regresses on the mixed training targets. This motivates transferring the physical update relative to its matched replay continuation, as described below.

Teacher preparation uses 6,232 raw energy/force queries: 702 at anchors and 5,530 at supported particles. An unweighted escort could omit the particle-energy queries; replay could omit all physical queries. Evaluating the four generated-output arms uses another 10,240 queries, reported separately from generation cost.

Transferring the paired physical update. We use parameter arithmetic (Ilharco et al., 2023) to transfer the difference between physical and replay continuations from the same initialization:

$$\theta_{\text{new}} = \theta_{\text{frozen}} + (\theta_{\text{physical}} - \theta_{\text{replay}}), \quad (16)$$

Table 8: Paired physical updates on the development panel. Energy differences are relative to the frozen model on common graph support, with paired 95% confidence intervals. This comparison follows the direct-distillation study.

Model	Graph passes (two streams)	Energy change (eV/atom)
Frozen	418/640; 377/640	0
Escort difference	407/640; 387/640	-0.01554 [-0.02084, -0.01031]
Work difference	406/640; 382/640	-0.01604 [-0.02188, -0.01013]

for both unweighted-escort and complete-work students. The coefficient is fixed at one. Parameter tensors are updated, while buffers and the source distribution remain fixed.

Both physical-update variants lower energy and force in both continuations. The work-difference variant reduces force RMS by 0.38403 eV/Å on common graph support (interval [-0.48124, -0.28932]); all-attempt energy and force readouts also improve. Its observed graph-rate difference is -0.55 percentage points, with interval [-2.58, 1.56].

The two physical-update variants have similar scores: their common-supported energy difference is -0.00243 eV/atom, with interval [-0.00756, 0.00264], while all-attempt energy difference is +0.00022 with interval [-0.00557, 0.00649]. Force and graph intervals also span zero. The unweighted variant requires 702 anchor queries for teacher preparation, compared with 6,232 for complete work. Both retain the same 128-call neural inference procedure.

Evaluating the two transferred variants adds 2,560 generated outputs and 5,120 physical queries. The parameter transfer itself requires no optimizer steps. We use these development results to choose the force-only update for the new-composition evaluation in Section 6.1.

H MOLECULAR ROTOR WORK PROTOCOL

The three training molecules are C=C(N)N(C)C(N)=O, CCOC(C)(C)N, and CC(=O)NCC=O. Selection examines the first five rows of an existing training subset, accepts the first three distinct compositions with a methyl rotor preserving the labeled perceived graph on a 64-angle grid, and queries no energy during selection. Only methyl hydrogens rotate around the carbon-anchor axis; the remaining internal coordinates stay fixed. Overall translation is removed. The reference and target densities are defined with respect to the periodic angle measure $d\phi$ on $[-\pi, \pi)$.

For $q_0(\phi) = \exp(\cos \phi)/(2\pi I_0(1))$, define reference Hamiltonian $H_0 = -kT \log q_0$ and target Hamiltonian $H_1 = H_0 + U - U(0)$. The escort has $|a| < 1$, so it is a periodic diffeomorphism. Its complete work is

$$W_a(\phi) = U(\psi_a(\phi)) - U(0) + kT [\log q_0(\phi) - \log q_0(\psi_a(\phi)) - \log J_a(\phi)]. \quad (17)$$

Change of variables gives

$$\mathbb{E}_{q_0} e^{-W_a/kT} = \int_{-\pi}^{\pi} q_0(y) e^{-[U(y)-U(0)]/kT} dy = Z_1/Z_0, \quad Z_0 = 1. \quad (18)$$

The logarithmic Jacobian accounts for the compression or expansion caused by the imposed escort.

All 1,536 single-point calculations succeed. The reference potential is a periodic cubic interpolant of 512 GFN2 values per rotor. Its 256-knot interpolation error at the additional midpoints is at most 10^{-8} eV in this study. Increasing the reference quadrature from 32,768 to 65,536 points changes the free energy by less than 10^{-10} eV. These checks quantify interpolation and quadrature error for the GFN2-based reference potential.

Total variation uses 64 equal angular bins, with reference probabilities obtained by dense quadrature. Self-normalized estimates of target observables have finite-sample bias; the mean unnormalized weight is an unbiased normalizer estimator. The reported free energy is $-kT \log \hat{Z}$, with its finite-sample bias included in RMSE. At 2,048 draws, complete-work ESS ranges from 788 to 1,892 across the six nonidentity cases. Some incomplete weights have higher ESS despite their biased target. Correct target recovery therefore depends on the work definition as well as the concentration of the weights.